Lecture 16

The Weyl character formula

Objectives. *A Verma character is an infinite series. Let $\lambda \in P_+$. We prove that the product of the Weyl denominator and the character of $L(\lambda)$ is a single finite alternant. The Casimir element puts the exponents of this product on a sphere. Dominance then leaves one Weyl orbit.*

By the end of this lecture you will be able to:

- Derive the Casimir scalar $(\mu, \mu + 2\rho)$ on a highest-weight module of highest weight μ.
- Write $\operatorname{ch} L(\lambda)$ as a finite integer combination of Verma characters.
- Prove the Weyl character formula for $\lambda \in P_+$ from the norm and dominance conditions.
- Display the cancellations for $\mathfrak{sl}_2$ and for the adjoint module of $\mathfrak{sl}_3$.

Throughout, $\mathfrak{g}$ is a nonzero complex semisimple Lie algebra with the data $\mathfrak{h}$, R_+, $\alpha_1, \dots, \alpha_r$, W, Q, P, P_+ and ρ of Lecture 9. Highest-weight modules are infinite-dimensional in general. This is a departure from the standing convention that modules are finite-dimensional. The final theorem concerns only the finite-dimensional modules $L(\lambda)$, $\lambda \in P_+$. We write

$$Q_+ = \sum_{i=1}^{r} \mathbb{Z}_{\geq 0}\alpha_i, \qquad \mu \leq \nu \iff \nu - \mu \in Q_+.$$

Characters lie in the completion of Lecture 15. Its elements are formal sums $\sum c_\nu e^\nu$ supported in a finite union of sets $\xi - Q_+$ with $\xi \in P$. For $\nu \in P$ define the alternant A_ν and the Weyl denominator D:

$$A_\nu = \sum_{w \in W} \det(w) e^{w\nu}, \qquad D = e^\rho \prod_{\alpha \in R_+} (1 - e^{-\alpha}).$$

From Lecture 15 we use the Weyl invariance of finite-dimensional characters and the Verma character: for $\mu \in P$,

$$\operatorname{ch} M(\mu) \prod_{\alpha \in R_+} (1 - e^{-\alpha}) = e^\mu.$$

Multiplying this identity by e^ρ gives

$$D \operatorname{ch} M(\mu) = e^{\mu + \rho}. \tag{16.1}$$

From Lecture 15 we also use the denominator identity $D = A_\rho$.

For $w \in W$, reindex by $u' = wu$. Then $\det(u) = \det(w)\det(u')$, so

$$\begin{aligned} wA_\nu &= \sum_{u \in W} \det(u) e^{wu\nu} = \det(w) \sum_{u' \in W} \det(u') e^{u'\nu} = \det(w) A_\nu, \\ wD &= \det(w) D. \end{aligned} \tag{16.2}$$

The last identity uses $D = A_\rho$.

16.1 The Casimir element of $U(\mathfrak{g})$

Fix the invariant symmetric form $(\cdot,\cdot)$ on $\mathfrak{g}$ of the course. This form is nondegenerate on $\mathfrak{g}$. On each simple ideal it is normalised so that long roots have squared length 2. Distinct simple ideals are orthogonal. On $\mathfrak{sl}_n$ it is $\operatorname{tr}(XY)$, while the Killing form is $2n \operatorname{tr}(XY)$.

The restriction of the form to $\mathfrak{h}$ is nondegenerate (Lecture 7). Hence for $\mu \in \mathfrak{h}^*$ there is a unique $t_\mu \in \mathfrak{h}$ with

$$(t_\mu, H) = \mu(H) \quad \text{for all } H \in \mathfrak{h}.$$

On $\mathfrak{h}^*$ put

$$(\mu, \nu) = (t_\mu, t_\nu) = \mu(t_\nu), \qquad \|\mu\|^2 = (\mu, \mu).$$

On the real span of the roots this form is W-invariant and positive definite (Lecture 8, Lecture 9).

Lemma 16.1 (Casimir element)**.** *Let (x_i) be a basis of $\mathfrak{g}$ and (x^i) the dual basis, so that $(x_i, x^j) = \delta_{ij}$. Then the element*

$$\Omega = \sum_i x_i x^i \in U(\mathfrak{g})$$

is central in $U(\mathfrak{g})$.

Proof. Fix $y \in \mathfrak{g}$. Expanding in the two bases gives

$$[y, x_i] = \sum_j c_{ji} x_j, \qquad c_{ji} = ([y, x_i], x^j).$$

The coefficient of x^j in $[y, x^i]$ is $([y, x^i], x_j)$. Invariance gives

$$([y, x^i], x_j) = -(x^i, [y, x_j]) = -\Big(x^i, \sum_k c_{kj} x_k\Big) = -c_{ij},$$

so $[y, x^i] = -\sum_j c_{ij} x^j$. In $U(\mathfrak{g})$, the map $u \mapsto yu - uy$ obeys the Leibniz rule. Hence

$$\begin{aligned} y\Omega - \Omega y &= \sum_i \big([y, x_i] x^i + x_i [y, x^i] \big) \\ &= \sum_{i,j} c_{ji} x_j x^i - \sum_{i,j} c_{ij} x_i x^j = 0. \end{aligned}$$

The last step exchanges i and j in the second sum. Since $\mathfrak{g}$ generates $U(\mathfrak{g})$, the element Ω is central. ∎

The change-of-basis calculation of Lecture 5, followed by multiplication into $U(\mathfrak{g})$, shows that Ω is basis-independent. Now choose a basis adapted to the root space decomposition. Root spaces are one-dimensional (Lecture 8). The form pairs $\mathfrak{g}_\alpha$ with $\mathfrak{g}_{-\alpha}$ nondegenerately.

It vanishes on all other pairs of root spaces and between $\mathfrak{h}$ and root spaces (Lecture 7). For $\alpha \in R_+$ choose $e_\alpha \in \mathfrak{g}_\alpha$ and $f_\alpha \in \mathfrak{g}_{-\alpha}$ with $(e_\alpha, f_\alpha) = 1$. Choose an orthonormal basis $H_1, \dots, H_r$ of $\mathfrak{h}$. It exists because the form on $\mathfrak{h}$ is symmetric and nondegenerate and the field is $\mathbb{C}$. The dual basis exchanges e_α and f_α and fixes each H_i. Hence

$$\Omega = \sum_{i=1}^{r} H_i^2 + \sum_{\alpha \in R_+} (e_\alpha f_\alpha + f_\alpha e_\alpha).$$

The bracket $[e_\alpha, f_\alpha]$ lies in $\mathfrak{g}_0 = \mathfrak{h}$. For $H \in \mathfrak{h}$,

$$(H, [e_\alpha, f_\alpha]) = ([H, e_\alpha], f_\alpha) = \alpha(H)(e_\alpha, f_\alpha) = \alpha(H).$$

Thus $[e_\alpha, f_\alpha] = t_\alpha$, which gives the first identity below. The second uses $\sum_{\alpha \in R_+} \alpha = 2\rho$ and the linearity of $\mu \mapsto t_\mu$:

$$e_\alpha f_\alpha = f_\alpha e_\alpha + t_\alpha, \qquad \sum_{\alpha \in R_+} t_\alpha = t_{2\rho}.$$

Substituting both identities into Ω gives

$$\Omega = \sum_{i=1}^{r} H_i^2 + t_{2\rho} + 2 \sum_{\alpha \in R_+} f_\alpha e_\alpha. \tag{16.3}$$

Definition 16.2 (Highest-weight module)**.** Let $\mu \in \mathfrak{h}^*$. A $\mathfrak{g}$-module U is a *highest-weight module of highest weight* μ if it is generated by a vector $v \neq 0$ with

$$e_\alpha v = 0 \ (\alpha \in R_+), \qquad Hv = \mu(H)v \ (H \in \mathfrak{h}).$$

Use $\mathfrak{n}_-$ from Lecture 11. By Lecture 12, such a module satisfies $U = U(\mathfrak{n}_-)v$. The module U is the direct sum of finite-dimensional weight spaces U_ν with $\nu \leq \mu$, and $U_\mu = \mathbb{C}v$. The Verma module $M(\mu)$ is one example. An irreducible highest-weight module of highest weight μ is isomorphic to $L(\mu)$.

Proposition 16.3 (Casimir scalar)**.** *On every highest-weight module U of highest weight μ,*

$$\Omega = (\mu, \mu + 2\rho)\ \mathrm{id}_U = \big(\|\mu + \rho\|^2 - \|\rho\|^2\big)\ \mathrm{id}_U\,.$$

Proof. Expand t_μ in the orthonormal basis:

$$t_\mu = \sum_i (t_\mu, H_i) H_i = \sum_i \mu(H_i) H_i, \qquad (\mu, \mu) = \mu(t_\mu) = \sum_i \mu(H_i)^2.$$

Apply (16.3) to v. Each e_α kills v, so the last sum vanishes:

$$\Omega v = \Big(\sum_i \mu(H_i)^2 + \mu(t_{2\rho})\Big) v = \big((\mu, \mu) + (\mu, 2\rho)\big) v = (\mu, \mu + 2\rho) v.$$

Every vector of U has the form uv with $u \in U(\mathfrak{g})$. Centrality gives

$$\Omega(uv) = u\,\Omega v = (\mu, \mu + 2\rho)\, uv.$$

∎

Replacing the form by $c \neq 0$ times itself divides Ω and its scalars by c. Exercise 16.1 compares the trace and Killing forms.

16.2 Highest-weight modules on one Casimir sphere

Lemma 16.4 (Weight submodules). *Let $V = \bigoplus_{\nu \in P} V_\nu$ be a $\mathfrak{g}$-module with finite-dimensional weight spaces. Suppose all weights lie in a finite union of sets $\xi - Q_+$ with $\xi \in P$. Highest-weight modules with highest weight in P are examples. Let $X \subseteq V$ be a submodule. Then $X = \bigoplus_\nu (X \cap V_\nu)$. Moreover, X and V/X satisfy the same hypotheses, so all three characters are defined, and*

$$\operatorname{ch} V = \operatorname{ch} X + \operatorname{ch}(V/X).$$

Proof. Let $u = v_1 + \cdots + v_k \in X$ with nonzero $v_j \in V_{\nu_j}$ and distinct ν_j. For $i \neq j$, the kernel of $\nu_i - \nu_j$ is a proper subspace of $\mathfrak{h}$. A complex vector space is not a finite union of proper subspaces. So some $H \in \mathfrak{h}$ has pairwise distinct values $\nu_j(H)$. Then

$$\prod_{j=2}^{k} \big(H - \nu_j(H)\big) u = \prod_{j=2}^{k} \big(\nu_1(H) - \nu_j(H)\big) v_1 \in X.$$

The scalar on the right is nonzero, so $v_1 \in X$. Likewise every v_j lies in X.

Hence V/X is the direct sum of the images of the V_ν. With $X_\nu = X \cap V_\nu$, the image of V_ν is V_ν/X_ν. It has weight ν, and

$$\dim (V/X)_\nu = \dim V_\nu - \dim X_\nu.$$

This is the character identity, coefficient by coefficient. ∎

Fix $\lambda \in P$ and define the *Casimir sphere*

$$S_\lambda = \{\mu \in P : \|\mu + \rho\|^2 = \|\lambda + \rho\|^2\}.$$

The basis $\omega_1, \dots, \omega_r$ makes P a lattice in a Euclidean space. A sphere contains only finitely many lattice points, so S_λ is finite. Subtracting $\|\rho\|^2$ from both sides of the defining condition gives

$$\mu \in S_\lambda \iff \mu \in P \text{ and } (\mu, \mu + 2\rho) = (\lambda, \lambda + 2\rho).$$

For $\mu \in P$, Proposition 16.3 gives $\mu \in S_\lambda$ exactly when Ω has the same scalar on $M(\mu)$ as on $M(\lambda)$.

The next proposition splits a highest-weight module by repeatedly passing to a submodule and a quotient. These modules may be infinite-dimensional, so we need a finite measure to show that the process stops. The Casimir element restricts every new highest weight to the finite set S_λ. We therefore count the dimensions of the weight spaces at those points.

For a highest-weight module U with highest weight in S_λ, set

$$N(U) = \sum_{\nu \in S_\lambda} \dim U_\nu.$$

It is finite, and $N(U) \geq 1$ because the highest weight contributes. For the submodule and quotient chosen in the proof below, this count splits into two positive parts, so the induction terminates.

Proposition 16.5 (Decomposition on the sphere)**.** *Let U be a highest-weight module of highest weight $\mu \in S_\lambda$. Then*

$$\operatorname{ch} U = \sum_{\nu \in S_\lambda,\ \nu \le \mu} a_\nu \operatorname{ch} L(\nu), \qquad a_\nu \in \mathbb{Z}_{\ge 0}, \quad a_\mu = 1.$$

Proof. We use strong induction on $N(U)$. If U is irreducible, then $U \cong L(\mu)$ and there is nothing to prove.

Step 1: a new highest-weight vector. Let X be a proper nonzero submodule. By Lemma 16.4, X is a sum of weight spaces. Also $X_\mu = 0$. Otherwise X would contain $U_\mu = \mathbb{C}v$, and then $X = U$. Pick a weight ν_0 of X. Consider the set

$$E = \{\xi : X_\xi \neq 0,\ \nu_0 \le \xi \le \mu\}.$$

It contains ν_0. It is finite, since $\xi - \nu_0 \in Q_+$ and only finitely many elements of Q_+ lie below $\mu - \nu_0$. Let η be maximal in E. All weights of U are $\le \mu$, so every weight $\xi > \eta$ of X would lie in E. Hence η is maximal among all weights of X. Also $\eta < \mu$, because $X_\mu = 0$. Take a nonzero vector $x \in X_\eta$. By the maximality of η, for every $\alpha \in R_+$,

$$e_\alpha x \in X_{\eta+\alpha} = 0.$$

Step 2: the Casimir. Since x is a weight vector annihilated by all positive root vectors, the calculation of Proposition 16.3 applies to x. Also $x \in U$, so

$$(\eta, \eta + 2\rho)x = \Omega x = (\mu, \mu + 2\rho)x.$$

As $\eta \in \mu - Q \subseteq P$, this gives $\eta \in S_\lambda$.

Step 3: induction. The submodule $X' = U(\mathfrak{g})x$ is a highest-weight module of highest weight η, and $X'_\mu = 0$ since $X' \subseteq X$. The quotient U/X' is a highest-weight module of highest weight μ, generated by the image of v. Lemma 16.4 gives

$$\operatorname{ch} U = \operatorname{ch} X' + \operatorname{ch}(U/X'), \qquad N(U) = N(X') + N(U/X').$$

Both summands of $N(U)$ are at least 1, so both are smaller than $N(U)$. By induction, $\operatorname{ch} X'$ involves only $L(\nu)$ with $\nu \le \eta < \mu$, and $\operatorname{ch}(U/X')$ contains $\operatorname{ch} L(\mu)$ exactly once. Adding the two expansions proves the claim. ∎

16.3 Triangular inversion and candidate weights

Enumerate $S_\lambda = \{\mu_1, \dots, \mu_l\}$ so that $\mu_i < \mu_j$ implies $i < j$. To do this, repeatedly remove a minimal element. Apply Proposition 16.5 to $U = M(\mu_k)$:

$$\operatorname{ch} M(\mu_k) = \operatorname{ch} L(\mu_k) + \sum_{j<k} a_{kj} \operatorname{ch} L(\mu_j). \tag{16.4}$$

Set $a_{kk} = 1$ and $a_{kj} = 0$ for $j > k$. Then $A = (a_{kj})$ is lower unitriangular with integer entries.

Proposition 16.6 (Inversion)**.** *For each k there are integers b_{kj} with $b_{kk} = 1$ and*

$$\operatorname{ch} L(\mu_k) = \sum_{j \le k} b_{kj} \operatorname{ch} M(\mu_j).$$

Proof. Use induction on k. Rearranging (16.4) gives

$$\operatorname{ch} L(\mu_k) = \operatorname{ch} M(\mu_k) - \sum_{j<k} a_{kj} \operatorname{ch} L(\mu_j).$$

By induction, each $\operatorname{ch} L(\mu_j)$ with $j < k$ is an integer combination of $\operatorname{ch} M(\mu_i)$ with $i \le j$. Substituting proves the claim. ∎

Theorem 16.7 (Finite candidate weights)**.** *Let $\lambda \in P_+$. There are integers b_μ such that*

$$D \operatorname{ch} L(\lambda) = \sum_{\mu \in S_\lambda} b_\mu e^{\mu+\rho}.$$

Every exponent ν with nonzero coefficient satisfies $\|\nu\|^2 = \|\lambda+\rho\|^2$.

Proof. Since $\lambda \in S_\lambda$, Proposition 16.6 writes $\operatorname{ch} L(\lambda)$ as an integer combination of $\operatorname{ch} M(\mu)$, $\mu \in S_\lambda$. Multiply by D and apply (16.1). The exponents $\mu+\rho$ are distinct and lie on the stated sphere. ∎

Both sides lie in $\mathbb{Z}[P]$, since $L(\lambda)$ is finite-dimensional (Lecture 13).

16.4 Eliminating lower alternants

Theorem 16.8 (Weyl character formula)**.** *Let $\lambda \in P_+$. Then $D \operatorname{ch} L(\lambda) = A_{\lambda+\rho}$. Since $D = A_\rho$,*

$$\operatorname{ch} L(\lambda) = \frac{A_{\lambda+\rho}}{A_\rho}, \qquad A_\nu = \sum_{w \in W} \det(w) e^{w\nu}.$$

Proof. Write $F = D \operatorname{ch} L(\lambda) = \sum_\nu c_\nu e^\nu$.

(a) Norm. If $c_\nu \neq 0$, then $\|\nu\|^2 = \|\lambda+\rho\|^2$, by Theorem 16.7.

(b) Dominance bound. Expand the product D:

$$D = e^\rho \prod_{\alpha \in R_+} (1 - e^{-\alpha}) = \sum_{T \subseteq R_+} (-1)^{|T|} e^{\rho - \sum_{\alpha \in T} \alpha}.$$

Each exponent of F therefore has the form

$$\nu = \mu + \rho - \sum_{\alpha \in T} \alpha, \qquad \mu \text{ a weight of } L(\lambda),\ T \subseteq R_+.$$

Since $\lambda - \mu$ and $\sum_{\alpha \in T} \alpha$ both lie in Q_+, every such ν satisfies $\nu \le \lambda + \rho$. Equality forces $\mu = \lambda$ and $T = \emptyset$, because a nonempty sum of positive roots is a nonzero element of Q_+. As $\dim L(\lambda)_\lambda = 1$, we get $c_{\lambda+\rho} = 1$.

(c) Alternation. The character is W-invariant (Lecture 15), and $wD = \det(w) D$ by (16.2). Since w acts on $\mathbb{Z}[P]$ by ring automorphisms,

$$wF = (wD)\,(w \operatorname{ch} L(\lambda)) = \det(w) D \operatorname{ch} L(\lambda) = \det(w) F.$$

Comparing coefficients of $e^{w\nu}$ gives $c_{w\nu} = \det(w) c_\nu$.

(d) Norm and dominance together. Let $c_\nu \neq 0$. By Lecture 9, $\nu' = w^{-1}\nu$ is dominant for some $w \in W$. By (c), $c_{\nu'} = \pm c_\nu \neq 0$, so (a) and (b) apply to ν'. By (b), the element $\gamma = \lambda + \rho - \nu'$ lies in Q_+. Then

$$\|\lambda+\rho\|^2 = \|\nu'+\gamma\|^2 = \|\nu'\|^2 + 2(\nu',\gamma) + \|\gamma\|^2,$$
$$0 = 2(\nu',\gamma) + \|\gamma\|^2.$$

The last line uses (a). Write $\gamma = \sum_i k_i \alpha_i$ with $k_i \geq 0$. Dominance gives $(\nu', \alpha_i) \geq 0$, hence

$$(\nu', \gamma) = \sum_i k_i (\nu', \alpha_i) \geq 0, \qquad \|\gamma\|^2 = -2(\nu', \gamma) \leq 0.$$

Positive definiteness forces $\gamma = 0$. Thus $\nu' = \lambda + \rho$ and $\nu = w(\lambda + \rho)$.

(e) Conclusion. By (b) and (c), $c_{w(\lambda+\rho)} = \det(w)$. If $w \neq 1$, some $\alpha \in R_+$ has $w\alpha \in -R_+$ (Lecture 9). For each simple root, $(\lambda, \alpha_i) \geq 0$ and $(\rho, \alpha_i) = \frac{1}{2}(\alpha_i, \alpha_i) > 0$. Every $\beta \in R_+$ is a nonzero combination of simple roots with nonnegative coefficients. Hence $(\lambda + \rho, \beta) > 0$ for all $\beta \in R_+$, and

$$(w(\lambda + \rho), w\alpha) = (\lambda + \rho, \alpha) > 0 > (\lambda + \rho, w\alpha).$$

So $w(\lambda + \rho) \neq \lambda + \rho$. Applying this to $u^{-1}w$ for $u \neq w$ shows that the points $w(\lambda + \rho)$ are distinct. By (d),

$$F = \sum_{w \in W} c_{w(\lambda+\rho)} e^{w(\lambda+\rho)} = \sum_{w \in W} \det(w) e^{w(\lambda+\rho)} = A_{\lambda+\rho}.$$

Finally, $\mathbb{Z}[P]$ is a Laurent polynomial ring in $e^{\omega_1}, \ldots, e^{\omega_r}$, hence an integral domain. By the expansion of D in (b), $A_\rho = D$ has top coefficient 1, so $A_\rho \neq 0$. The quotient is the unique element whose product with A_ρ is $A_{\lambda+\rho}$. ∎

16.5 Rank one and the adjoint module of $\mathfrak{sl}_3$

Example 16.9 ($\mathfrak{sl}_2$)**.** Here $\rho = \omega = \alpha/2$, $(\omega, \omega) = \frac{1}{2}$ and $W = \{1, s\}$. Write $x = e^\omega$ and $\lambda = m\omega$, with $m \in \mathbb{Z}_{\geq 0}$. For $\mu = k\omega$,

$$\|\mu + \rho\|^2 = \frac{1}{2}(k+1)^2, \qquad \frac{1}{2}(k+1)^2 = \frac{1}{2}(m+1)^2 \iff k \in \{m, -m-2\}.$$

Hence

$$S_\lambda = \{m\omega,\ -(m+2)\omega\}.$$

Exercise 16.2 proves the Verma expansion on this sphere:

$$\operatorname{ch} L(m\omega) = \operatorname{ch} M(m\omega) - \operatorname{ch} M(-(m+2)\omega).$$

Multiply by $D = A_\rho = x - x^{-1}$ and use (16.1):

$$A_\rho \operatorname{ch} L(m\omega) = e^{(m+1)\omega} - e^{-(m+1)\omega} = x^{m+1} - x^{-m-1}.$$

The cancellations are also visible directly. For $m = 0$ both sides equal $x - x^{-1}$. For $m \geq 1$,

$$\begin{aligned}(x - x^{-1}) \sum_{k=0}^{m} x^{m-2k} &= (x^{m+1} + x^{m-1} + \cdots + x^{-m+1}) \\ &\quad - (x^{m-1} + \cdots + x^{-m+1} + x^{-m-1}) \\ &= x^{m+1} - x^{-m-1}.\end{aligned}$$

The next example tests the proof on the adjoint module of $\mathfrak{sl}_3$. Figure 16.1 shows its two alternants $A_{2\rho}$ and A_ρ in the Euclidean plane of Lecture 9.

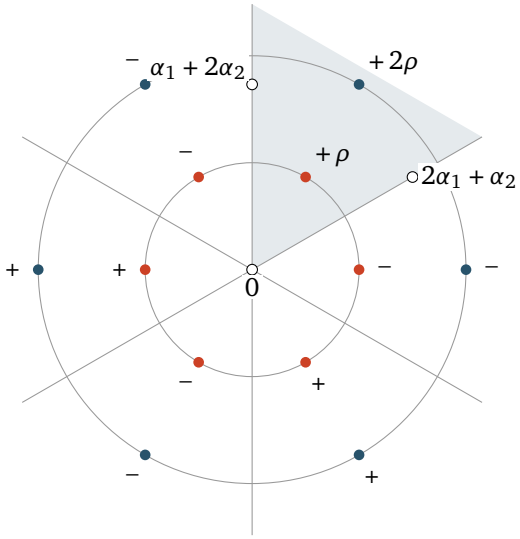


Figure 16.1: The two signed orbits of the example: $W(2\rho)$ on the circle of squared norm 8 and $W\rho$ on the circle of squared norm 2. The shaded region is the dominant chamber. The open dots are the other dominant exponents in the table of the example, whose coefficients vanish.

Example 16.10 (The adjoint module of $\mathfrak{sl}_3$)**.** Here $\theta = \alpha_1 + \alpha_2 = \rho$ and $\|\rho\|^2 = 2$. For $w = 1, s_1, s_2, s_1s_2, s_2s_1, w_0$, with $w_0 = s_1s_2s_1$, the points $w\rho$ are, in this order,

$$\alpha_1+\alpha_2, \quad \alpha_2, \quad \alpha_1, \quad -\alpha_1, \quad -\alpha_2, \quad -\alpha_1-\alpha_2.$$

Hence

$$A_\rho = e^{\alpha_1+\alpha_2} - e^{\alpha_2} - e^{\alpha_1} + e^{-\alpha_1} + e^{-\alpha_2} - e^{-\alpha_1-\alpha_2},$$
$$A_{2\rho} = e^{2\alpha_1+2\alpha_2} - e^{2\alpha_2} - e^{2\alpha_1} + e^{-2\alpha_1} + e^{-2\alpha_2} - e^{-2\alpha_1-2\alpha_2}.$$

The adjoint module is $L(\theta)$. Its nonzero weights are the six roots, each with multiplicity one. We assume only this and let the formula determine the multiplicity m of the weight 0. Set

$$\chi_m = \sum_{\alpha\in R} e^\alpha + m\,e^0.$$

Every χ_m is W-invariant, so $A_\rho\chi_m$ is alternating, as in step (c). Its coefficients on a W-orbit are fixed by the dominant representative. So it suffices to compute the dominant coefficients. Alternation determines the other coefficients with their Weyl signs. Every weight of χ_m is $\le \theta$. So the argument of (b), applied to $A_\rho\chi_m$, bounds all exponents by 2ρ. Since $\omega_1 = \frac{1}{3}(2\alpha_1+\alpha_2)$ and $\omega_2 = \frac{1}{3}(\alpha_1+2\alpha_2)$, the dominant elements of Q below 2ρ are those in the first column. The table shows which lower alternant remains:

Exponent	Products (term of A_ρ)·(term of χ_m)	Coefficient
$2\alpha_1+2\alpha_2$	$e^\rho\cdot e^\theta$	1
$2\alpha_1+\alpha_2$	$e^\rho\cdot e^{\alpha_1}$, $-e^{\alpha_1}\cdot e^\theta$	0
$\alpha_1+2\alpha_2$	$e^\rho\cdot e^{\alpha_2}$, $-e^{\alpha_2}\cdot e^\theta$	0
$\alpha_1+\alpha_2$	$e^\rho\cdot me^0$, $-e^{\alpha_1}\cdot e^{\alpha_2}$, $-e^{\alpha_2}\cdot e^{\alpha_1}$	$m-2$
0	$\det(w)e^{w\rho}\cdot e^{-w\rho}$, $w\in W$	$3-3=0$

The exponents $2\alpha_1+\alpha_2$, $\alpha_1+2\alpha_2$ and 0 lie on reflecting lines, so step (c) also forces their coefficients to vanish. Therefore

$$A_\rho \chi_m = A_{2\rho} + (m-2)A_\rho.$$

The surplus $(m-2)A_\rho$ is a lower alternant, on the inner circle of Figure 16.1. Alternation (c) and the top coefficient (b) cannot remove it. The norm condition (a) does. Using the Gram matrix in Lecture 9, for $\nu = a\omega_1 + b\omega_2$,

$$\|\nu\|^2 = \frac{2}{3}(a^2+ab+b^2), \qquad \|2\rho\|^2 = \frac{2}{3}(4+4+4) = 8.$$

The condition $\|\nu\|^2 = 8$ reads $a^2+ab+b^2 = 12$, while $\|\rho\|^2 = 2 \neq 8$. Theorem 16.8 therefore forces $m = 2$:

$$\dim L(\theta)_0 = 2 = \dim \mathfrak{h}, \qquad \dim L(\theta) = 6+2 = 8.$$

A Casimir sphere can contain more than one Weyl orbit, and Exercise 16.4 gives an example. In that example the dominance bound (b) removes the extra orbit. As a preview, Lecture 28 returns to this sphere. That lecture replaces the single element Ω by the whole centre of $U(\mathfrak{g})$. For highest weights λ and μ, equal central scalars then force $\mu+\rho \in W(\lambda+\rho)$. Nothing in this lecture depends on that result.

Self-checks

(1) What is the Casimir scalar on the adjoint module of $\mathfrak{sl}_3$, for $\operatorname{tr}(XY)$ and for the Killing form?

(2) For $\mathfrak{sl}_2$ and $\lambda = 3\omega$, what are S_λ and the Verma expansion of $\operatorname{ch} L(\lambda)$?

(3) Why does e^0 never occur in $A_\rho \operatorname{ch} L(\lambda)$?

Answers. (1) $(\theta, \theta+2\rho) = (\rho, 3\rho) = 6$. The Killing form is $6\operatorname{tr}(XY)$, so its scalar is 1. (2) $S_\lambda = \{3\omega, -5\omega\}$ and $\operatorname{ch} L(3\omega) = \operatorname{ch} M(3\omega) - \operatorname{ch} M(-5\omega)$. (3) Since $\mathfrak{g} \neq 0$, there is a simple reflection s_1. It fixes 0, so (c) gives $c_0 = -c_0$.

Summary

The Casimir element acts on a highest-weight module of highest weight μ by $\|\mu+\rho\|^2 - \|\rho\|^2$. The Verma characters with highest weights on a finite Casimir sphere are unitriangular combinations of irreducible characters. Let $\lambda \in P_+$. Inverting the triangular system puts the exponents of $A_\rho \operatorname{ch} L(\lambda)$ on the sphere. Dominance and the norm leave only $W(\lambda+\rho)$, and this gives $A_\rho \operatorname{ch} L(\lambda) = A_{\lambda+\rho}$. Lecture 17 turns the formula into dimensions and multiplicities.

Reading

B. Hall, *Lie Groups, Lie Algebras, and Representations*, second edition (2015), Sections 10.7–10.8; Hall uses a convex-hull argument where we use dominance. J. E. Humphreys, *Introduction to Lie Algebras and Representation Theory*, Section 24 and its appendix. R. Carter, *Lie Algebras of Finite and Affine Type* (2005), Chapter 12. Exercise Sheet 16 repeats the proof in rank one.